Общесос

Доп семинар — Главн. Физ. Четверг 6:30 PM

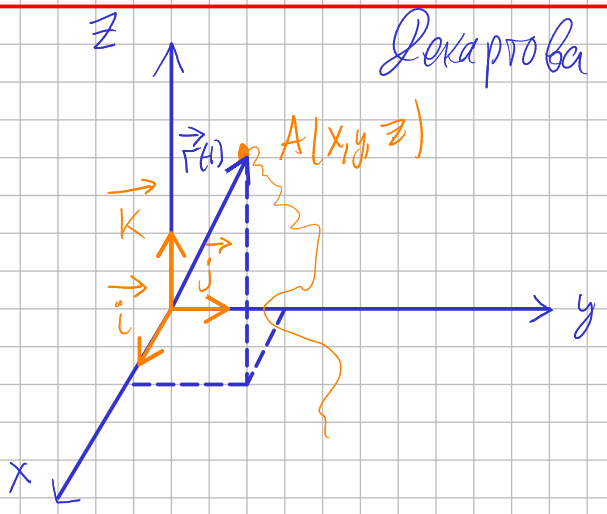
Кинематика

в Классической механике свойства пространства и времени не зависят от т.л.

К-а описывает движение, не рассматривая его причин.

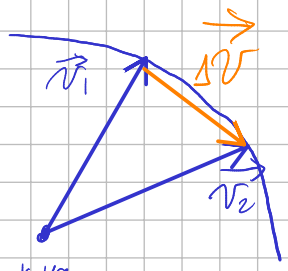
Мат. точка — точка тела, чьими размерами по сравнению с движением пренебречь

Декартова система координат

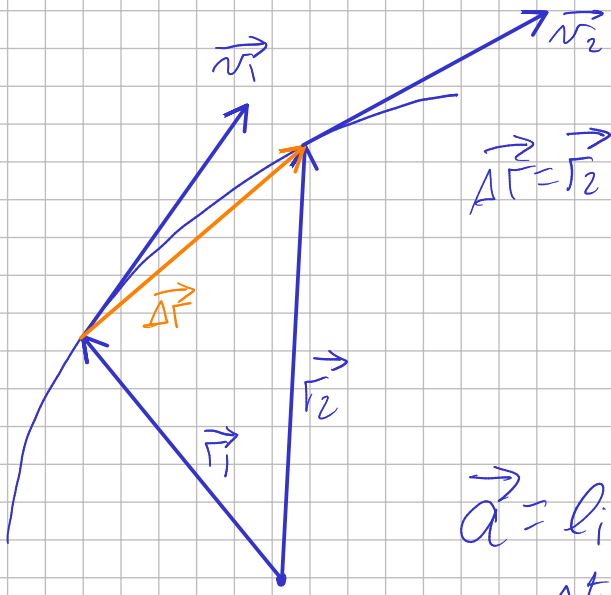


$$\vec{r}(t) = x(t)\cdot\vec{i} + y(t)\cdot\vec{j} + z(t)\cdot\vec{k}$$

$$|\vec{r}| = x^2+y^2+z^2$$



$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$



$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

Неподвижная точка

Траектория скоростей — годограф скоростей

$$\vec{a} = \lim_{\Delta t\to 0}\frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \dot{\vec{v}} = \frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}}$$

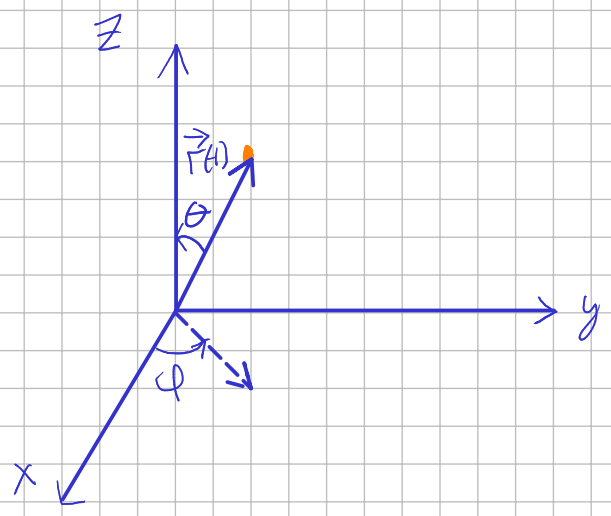
$$\vec{v}_{cp} = \frac{\Delta\vec{r}}{\Delta t}$$

$$\vec{v} = \lim_{\Delta t\to 0}\frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} = \dot{\vec{r}}$$

$$\vec{v}(t) = \underbrace{\frac{dx}{dt}}_{v_x}\cdot\vec{i} + \underbrace{\frac{dy}{dt}}_{v_y}\cdot\vec{j} + \underbrace{\frac{dz}{dt}}_{v_z}\cdot\vec{k}$$

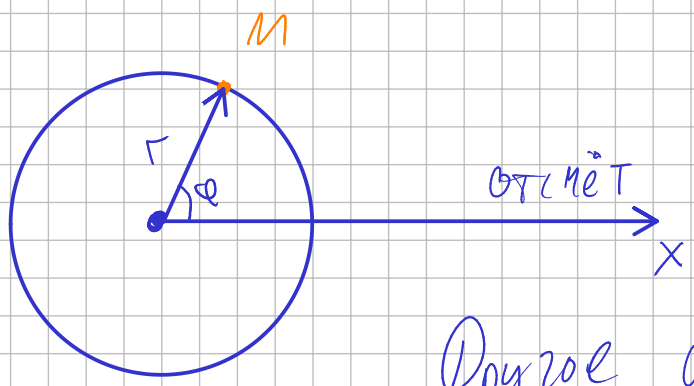
$$\vec{a}(t) = a_x\cdot\vec{i} + a_y\cdot\vec{j} + a_z\cdot\vec{k}$$

Полярная система координат



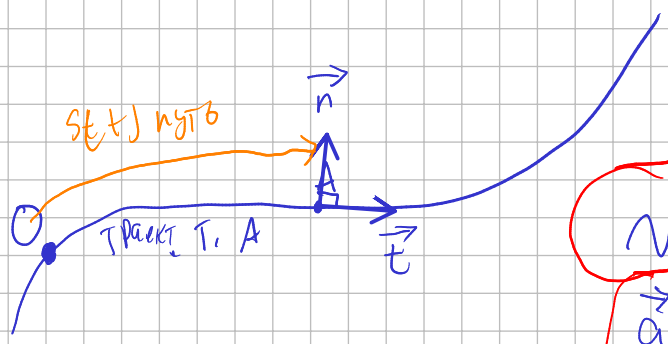
$$x = r\sin\theta\cos\varphi$$
$$y = r\sin\theta\sin\varphi$$
$$z = r\cos\theta$$

$$\varphi = \omega t \qquad T = \frac{1}{\nu}$$
$$v = \omega r$$
$$\nu = \frac{\omega}{2\pi} \qquad \dot{\omega} = \frac{d\omega}{dt} = \frac{d^2\varphi}{dt^2} = \ddot{\varphi}$$



Другое описание движения.

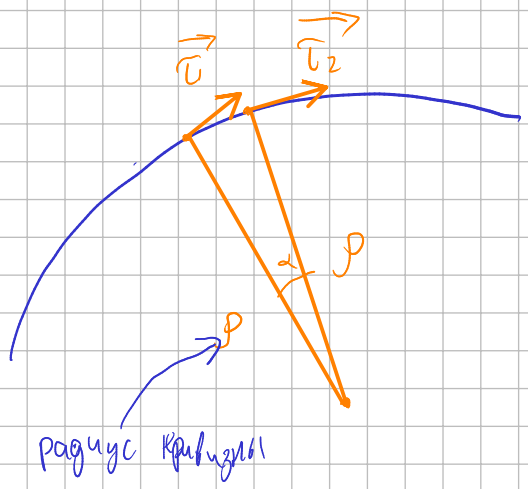
$$\vec{n}(t) \qquad \vec{v}(t)$$
$$|\vec{v}| = |\vec{n}| = 1$$



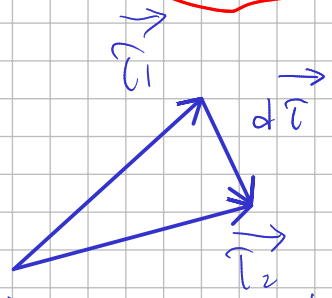
$$v = \frac{ds}{dt} = \dot{s} \qquad \vec{v} = v\vec{\tau} = \dot{s}\vec{\tau}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \underbrace{\frac{dv}{dt}}_{a_\tau}\vec{\tau} + \frac{v\,d\vec{\tau}}{dt} = a_\tau\vec{\tau} + \underbrace{\frac{v^2}{\rho}}_{a_n}\vec{n}$$

— ц.с. уск.



$$\alpha = \frac{ds}{\rho}$$



$$\boxed{a = \sqrt{a_\tau^2 + a_n^2}}$$

$$\alpha = \frac{|d\vec{\tau}|}{|\vec{\tau}| = 1} = \frac{ds}{\rho} \qquad \frac{|d\vec{\tau}|}{ds} = \frac{1}{\rho}$$

$$\frac{d\vec{\tau}}{ds} = \frac{1}{\rho}\vec{n}$$

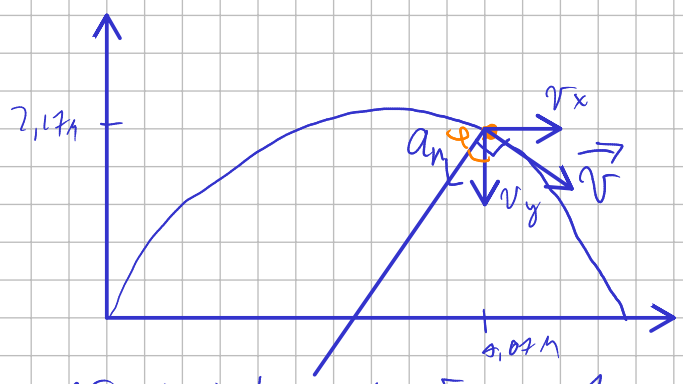
$$\frac{d\vec{\tau}}{dt} = \frac{d\vec{\tau}}{ds}\cdot\underbrace{\left(\frac{ds}{dt}\right)}_{v} = \frac{v\vec{n}}{\rho}$$

$v_0 = 10\ \frac{м}{с}$

$\alpha = 45^\circ$

$t = 1с$

$\rho = ?$



$$x(t) = v_0\cos\alpha\, t = 10\cdot\frac{\sqrt{2}}{2} \approx 7{,}07\ м$$

$$y(t) = v_0\sin\alpha\, t - \frac{gt^2}{2} = 10\cdot\frac{\sqrt{2}}{2}\cdot 1 - \frac{9{,}81\cdot 1}{2} = 2{,}17\ м$$

$$v_x(x) = v_0\cos\alpha = 7{,}07\ \frac{м}{с}$$

$$v_y(t) = v_0\sin\alpha - gt = 10\cdot\frac{\sqrt{2}}{2} - 9{,}81 = -2{,}74\ \frac{м}{с}$$

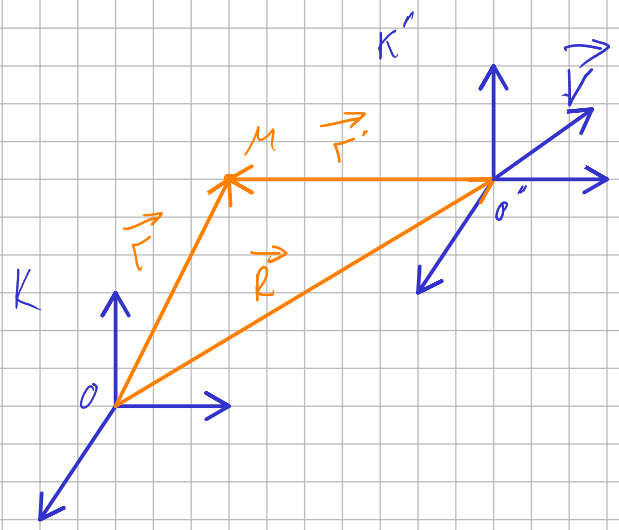
$$\cos\varphi = \frac{v_x}{v} = 0{,}933 \qquad \rho = \frac{v^2}{a_n} \qquad v = \sqrt{v_x^2 + v_y^2} = 7{,}58\ \frac{м}{с}$$

$$a_n = g\cos\varphi = 9{,}15\ \frac{м}{с^2}$$

$$\rho = \frac{7{,}58^2}{9{,}15} = 6{,}3\ м$$

Сивухин, Кириченко-Крымский

$$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}}$$



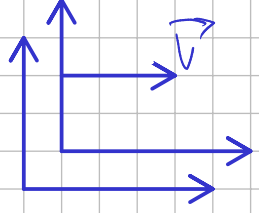
$$\vec{R} = \vec{V}t + \vec{R}_0$$

$$\vec{r} = \vec{r}' + \vec{R}$$

$$\dot{\vec{r}} = \dot{\vec{r}}' + \vec{V}$$

$$\ddot{\vec{r}} = \ddot{\vec{r}}', \text{ т.к.}$$

$$\boxed{\begin{aligned}\vec{v} &= \vec{v}' + \vec{V}\\ \vec{a} &= \vec{a}'\end{aligned}}$$



$$\left.\begin{aligned} x &= x' + vt \\ y &= y' \\ z &= z' \end{aligned}\right| \; t = t' \Bigg\} \text{ преобразования Галилея}$$

$\Downarrow$

Все законы механики ковариантны относительно преобразований Галилея.